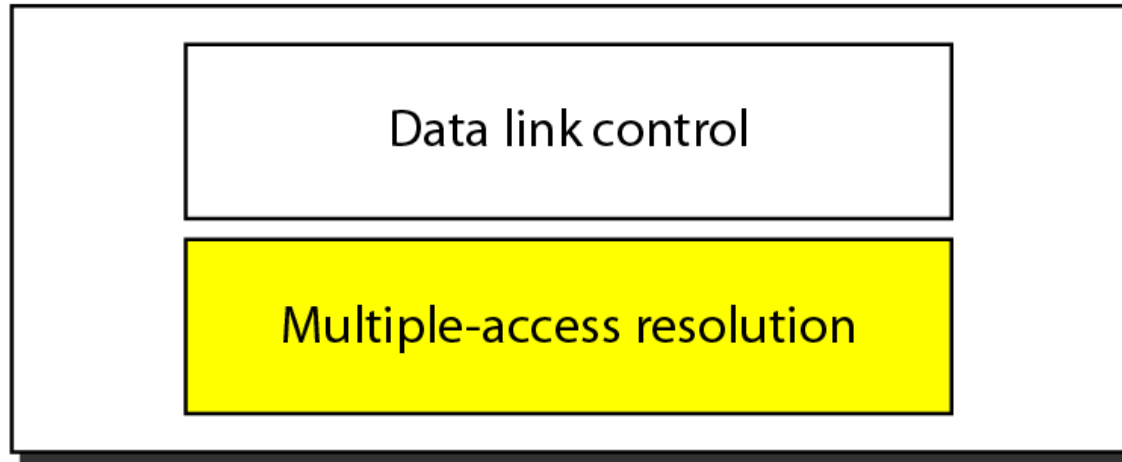


Multiple Access Control (MAC)

Data link layer divided into two functionality-oriented sublayers

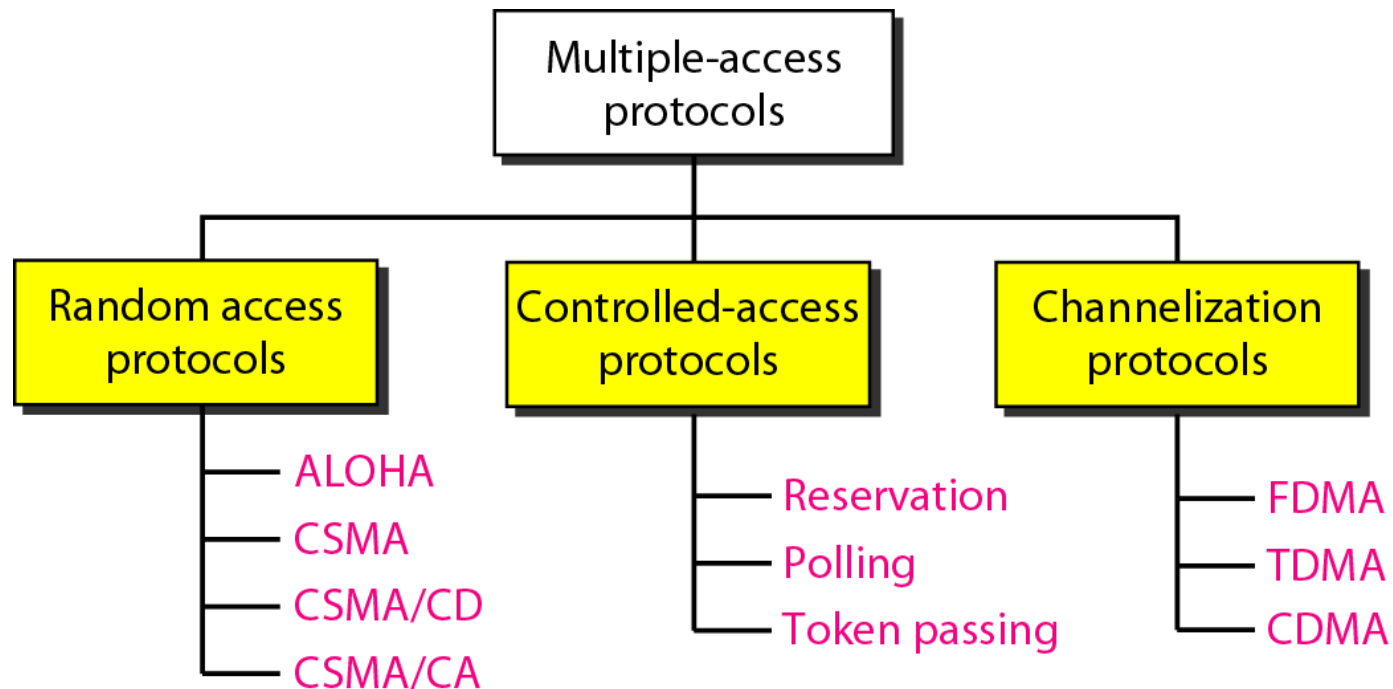
Data link layer



- Upper sublayer is responsible for encryption, flow and error control (**Logical Link Control**)
- Lower sublayer is responsible for resolving access to the shared medium (**Medium Access Control**)

Taxonomy of multiple-access protocols

- Two or more nodes are connected to a common (called a multipoint or broadcast) link, Medium Access Control protocol is used to coordinate the access to the link
- Many formal protocols were developed to handle access to a shared link



Random Access

- In random access or contention methods, no station is superior to another station and none is assigned the control over another.
- No station permits / does not permit, another station to send.
- Each station follows a predefined procedure, including testing of the medium, whenever it has data to send.
- The decision depends on the state of the medium (idle or busy).

Random Access

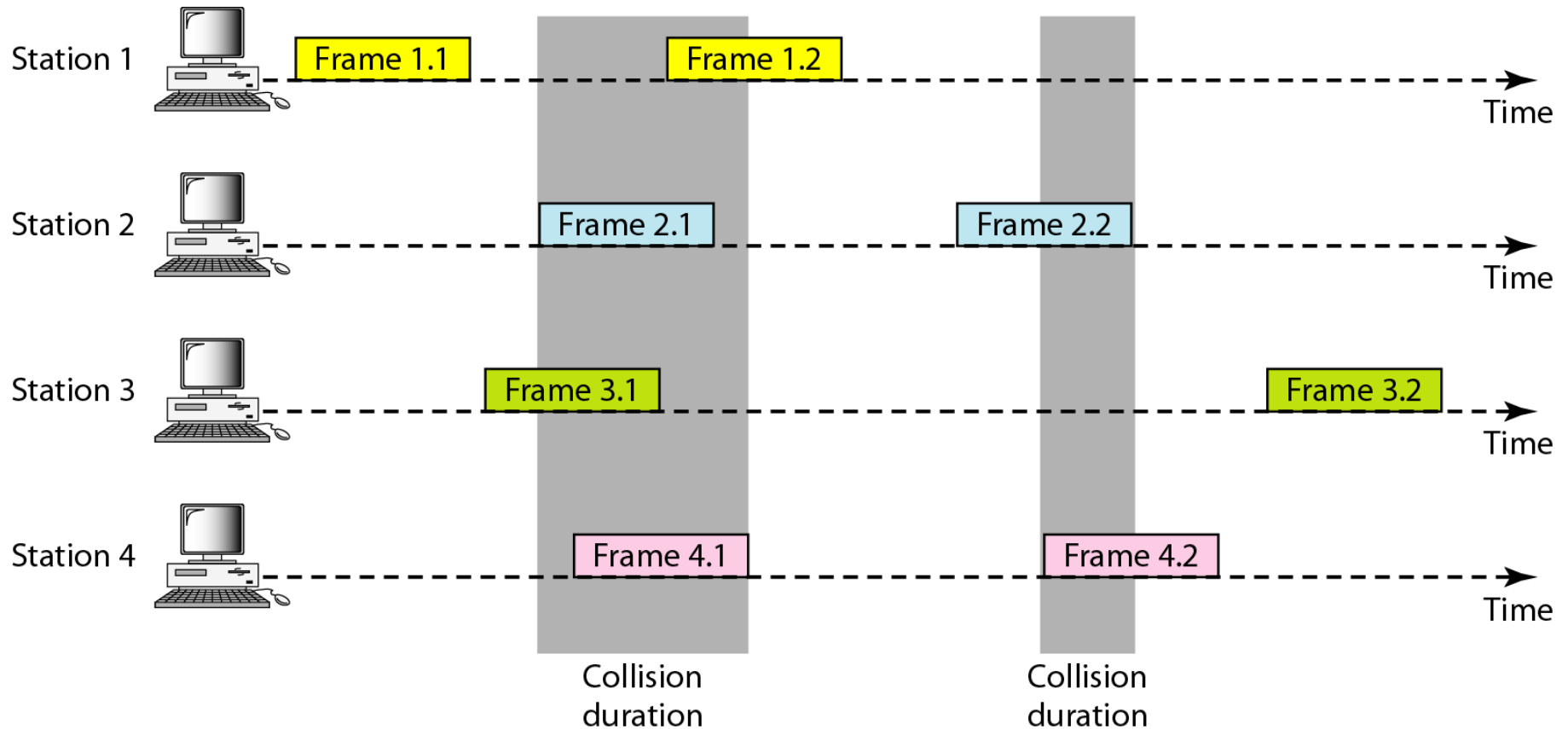
- No scheduled time for a station to transmit
- Transmission is random among the stations(Random Access)
- No rules specify which station should send next.
- Stations compete with one another to access the medium(Contention Methods)

- If more than one station tries to send, there is an access conflict called **collision**
 - Frames will be either destroyed or modified
- To avoid the access conflict, the following questions need to be answered
 - When can the station access the medium
 - What can the station do if the medium is busy
 - How can the station determine the success or failure of the transmission
 - What can the station do if there is an access conflict

ALOHA

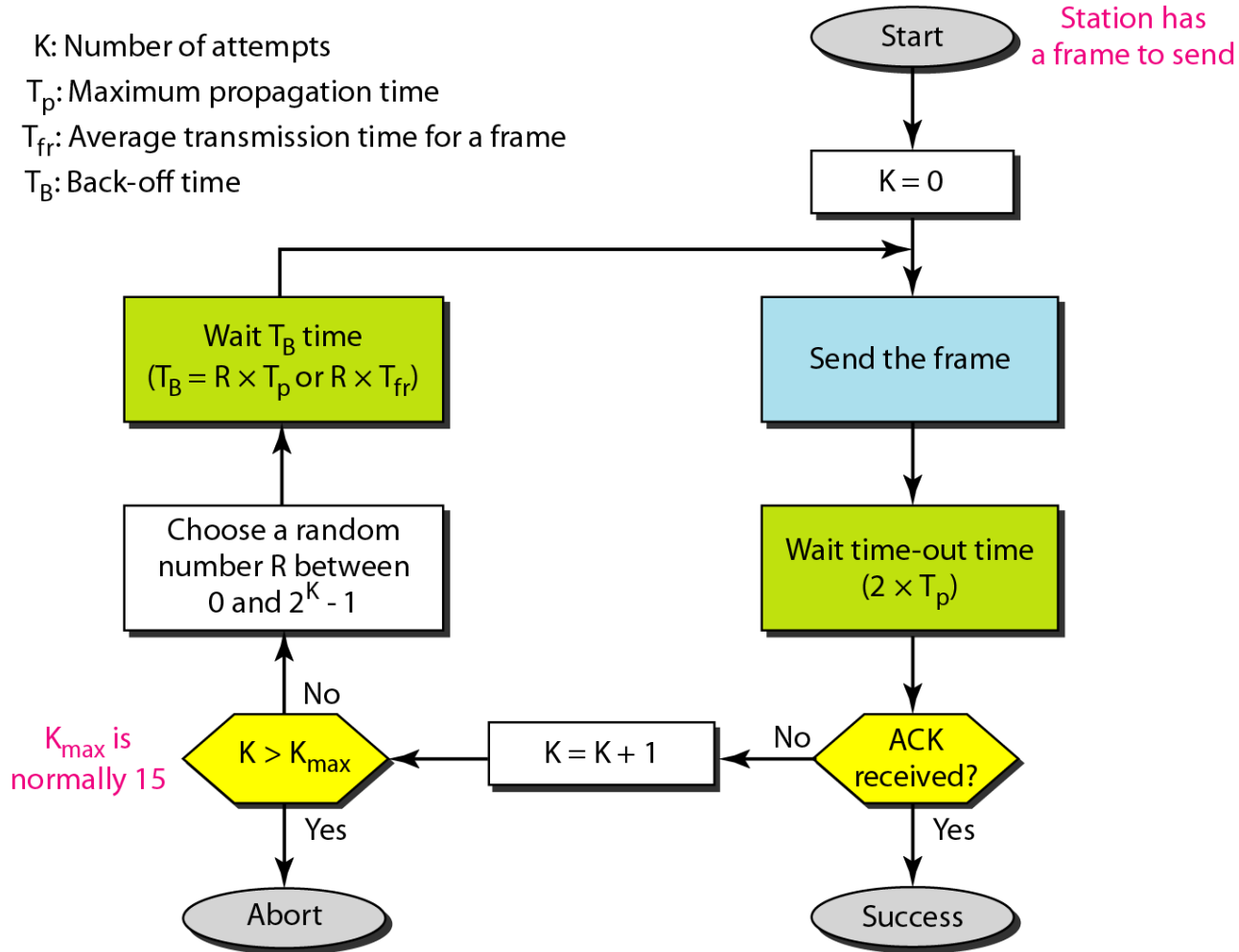
- ALOHA was developed at University of Hawaii, in 1970's
- Developed for wireless LAN but can be used for any shared medium
- Called pure ALOHA
- Idea:
 - Station can transmit the frame whenever it has frame to send

Frames in a pure ALOHA network

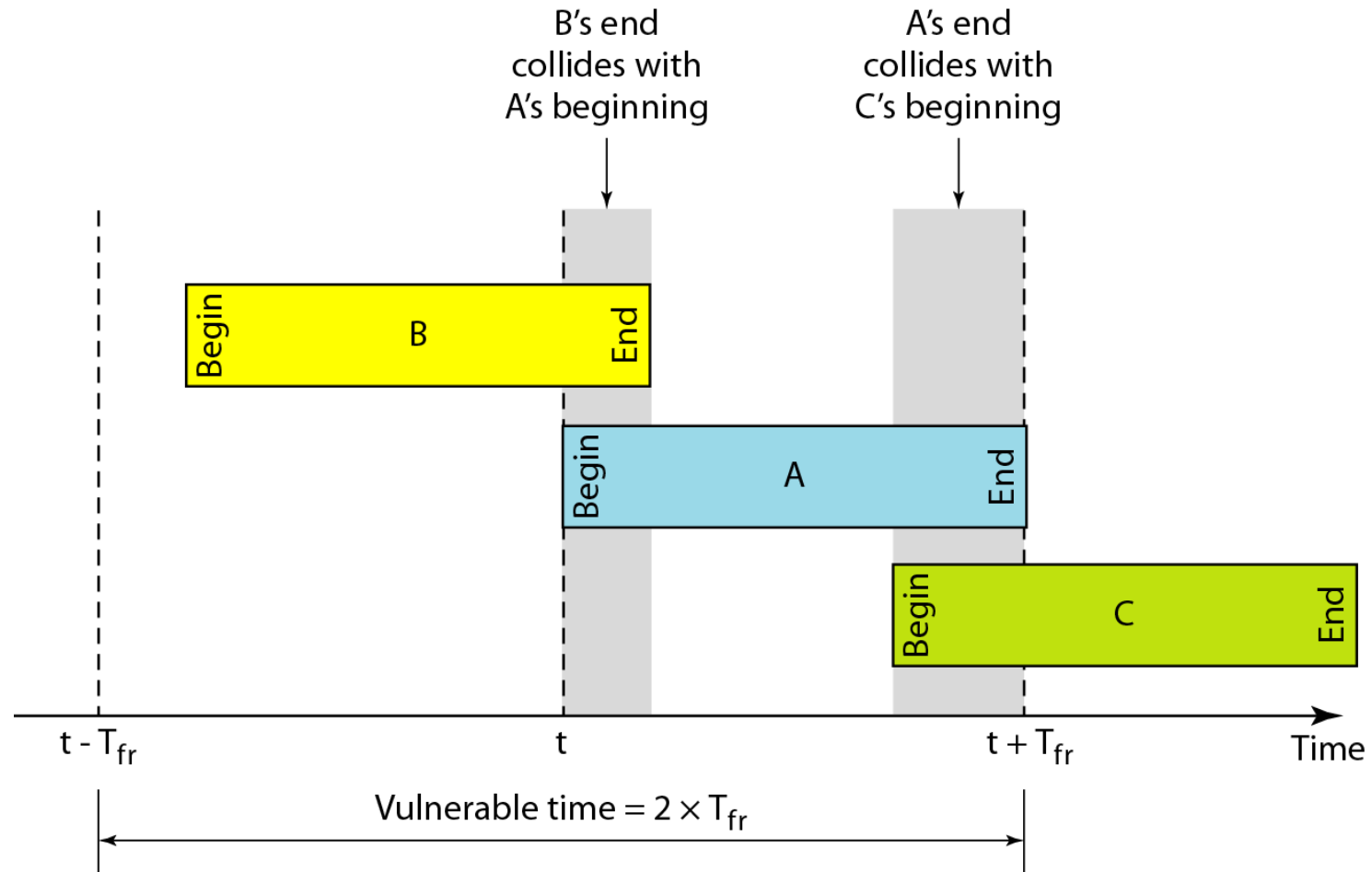


- Pure ALOHA protocol expects **acknowledgements from the receiver**
- If Timer expires, **sender needs to resend the frame**
- Stations resend the frame after timeout then again collision will occur
 - So each station needs to wait for another random time period (Backoff time)
- Maximum No. of retransmission attempts is $K_{\max}$.
- Timeout period is equal to **maximum possible round trip delay (2 * transmission delay)**
- Back off Time :- Random Wait time is R (b/w 0 and $2^k - 1$) * Max Propagation time or Average time required to send a frame.

Procedure for pure ALOHA protocol



Vulnerable time for pure ALOHA protocol



- Vulnerable time, Time period having the possibility of collision ($2 \times T_{fr}$)
- Fixed length frames, each taking T_{fr} to send

Example 12.2

A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the requirement to make this frame collision-free?

Solution

Average frame transmission time T_{fr} is 200 bits/200 kbps or 1 ms. The vulnerable time is $2 \times 1 \text{ ms} = 2 \text{ ms}$. This means no station should send later than 1 ms before this station starts transmission and no station should start sending during the period (1 ms) that this station is sending.

The throughput for pure ALOHA is

$$S = G \times e^{-2G} .$$

Where G is the average number of frames transmitted,
S is the average No. of successful transmissions.

Euler's number is 'e' is 2.71828

The maximum throughput

$$S_{\max} = 0.184 \text{ when } G = (1/2).$$

A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces

- a. 1000 frames per second?
- b. 500 frames per second?
- c. 250 frames per second?

Solution

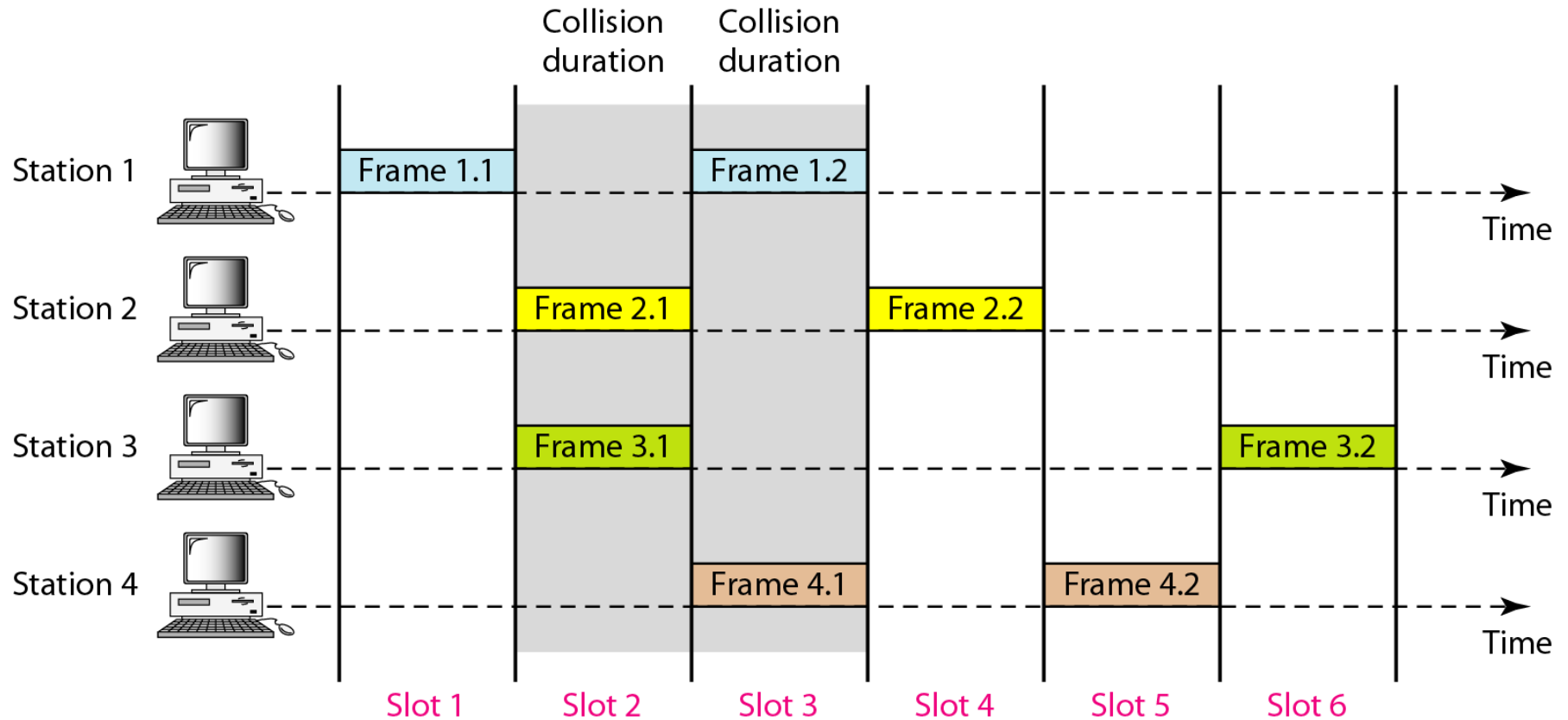
The frame transmission time is 200/200 kbps or 1 ms.

- a. If the system creates 1000 frames per second, or 1 frame per millisecond, then $G = 1$. In this case $S = G \times e^{-2G} = 0.135$ (13.5 percent). This means that the throughput is $1000 \times 0.135 = 135$ frames. Only 135 frames out of 1000 will probably survive.
- b. If the system creates 500 frames per second, or 1/2 frames per millisecond, then $G = 1/2$. In this case $S = G \times e^{-2G} = 0.184$ (18.4 percent). This means that the throughput is $500 \times 0.184 = 92$ and that only 92 frames out of 500 will probably survive. Note that this is the *maximum* throughput case. percentagewise.
- c. If the system creates 250 frames per second, or 1/4 frames per millisecond, then $G = 1/4$. In this case $S = G \times e^{-2G} = 0.152$ (15.2 percent). This means that the throughput is $250 \times 0.152 = 38$. Only 38 frames out of 250 will probably survive.

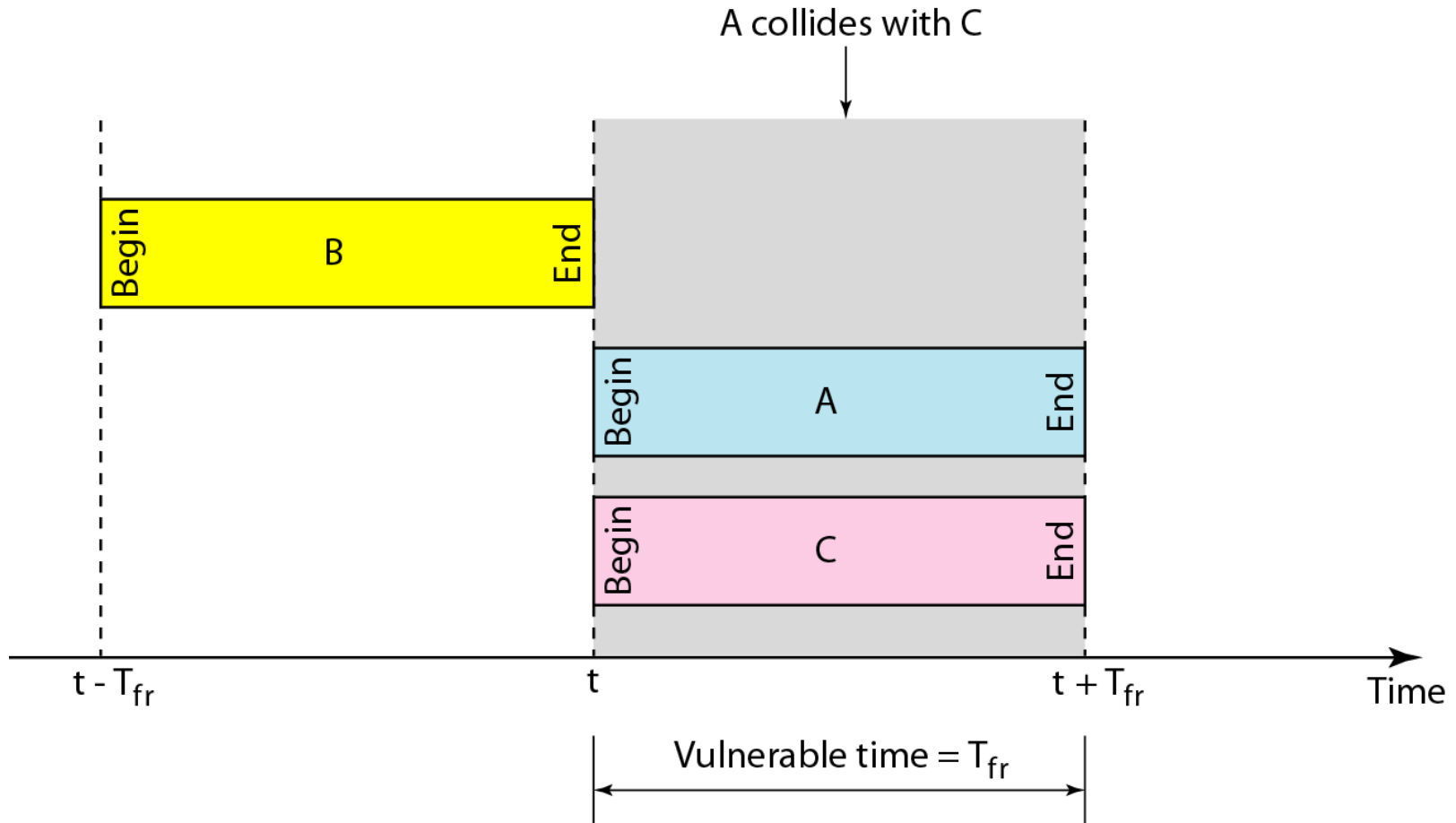
Slotted ALOHA

- Vulnerable time of pure ALOHA = $2 * T_{fr}$
- No rule to decide when the station should send the data
- Slotted ALOHA to improve the efficiency
- Forces the station to send the frame at the beginning of the time slot
- Station misses its chance should wait for the next time slot.

Frames in a slotted ALOHA network



Vulnerable time for slotted ALOHA protocol



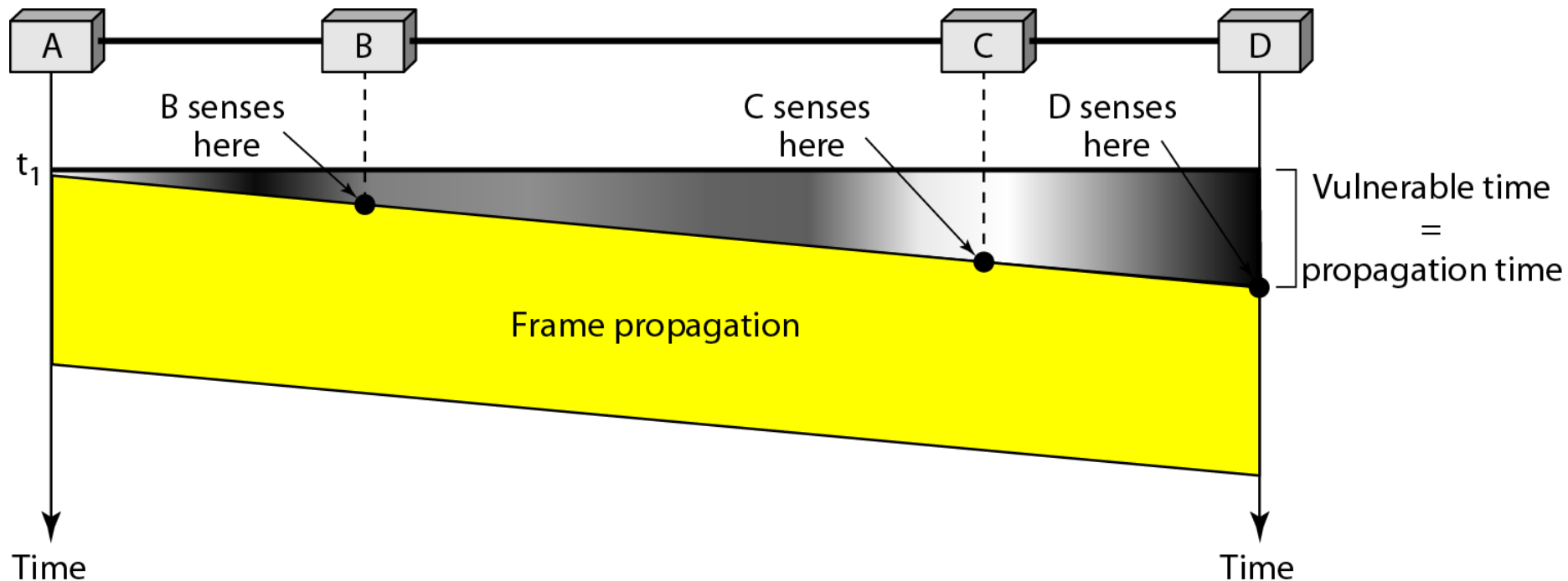
The throughput for slotted ALOHA is $S = G \times e^{-G}$.
The maximum throughput $S_{max} = 0.368$ when $G = 1$.

Carrier Sense Multiple Access (CSMA)

- CSMA was developed
 - To minimize the chance of collision
 - To increase the efficiency
- CSMA requires the station to first listen to the medium then transmit
- Collision can be reduced but can't be eliminated

Vulnerable time in CSMA

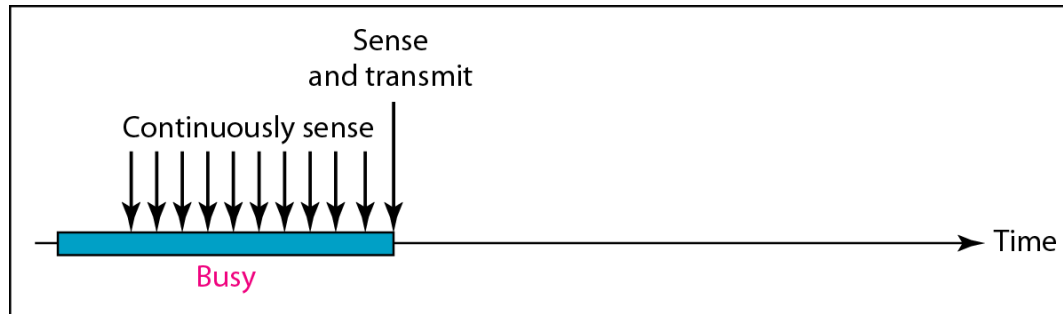
The Vulnerable time for CSMA is **Propagation time**



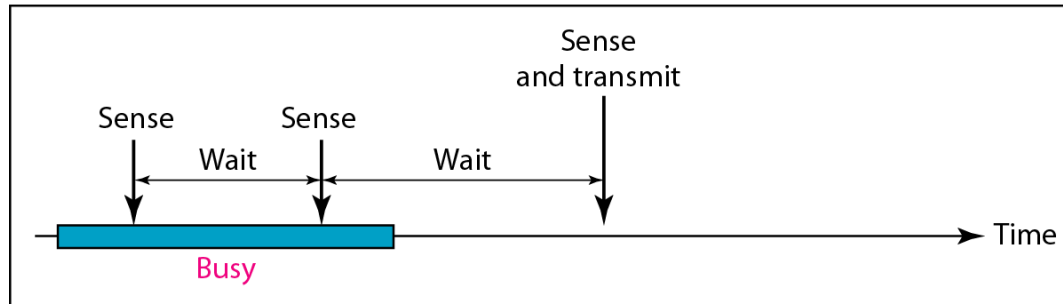
Persistence Methods

- What a station should do
 - If the channel is busy?
 - If the channel is idle?
- Persistence strategy defines the procedures for a station that senses a busy medium

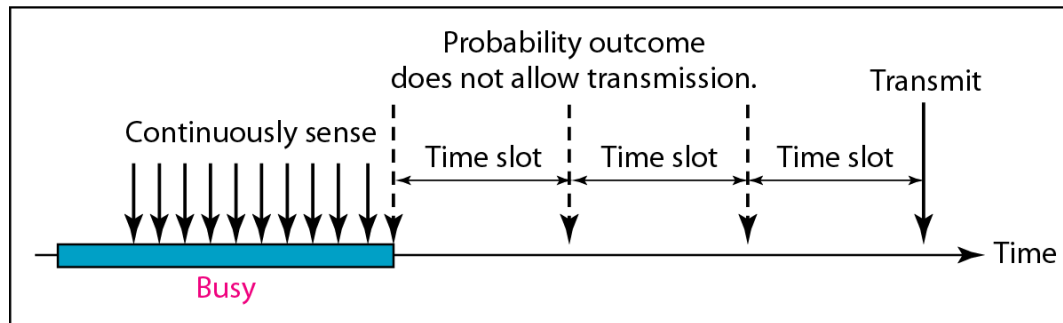
Behavior of three persistence methods



a. 1-persistent



b. Nonpersistent



c. p-persistent

1-Persistent CSMA

- In this method, station that wants to transmit data continuously sense the Channel to check whether the channel is idle or busy.
- If the channel is busy , the station waits until it becomes idle.
- When the station detects an idle channel, it immediately transmits the frame with probability 1. Hence it is called 1-persistent CSMA.
- highest chance of collision
- When the collision occurs, the stations wait a random amount of time and start all over again.

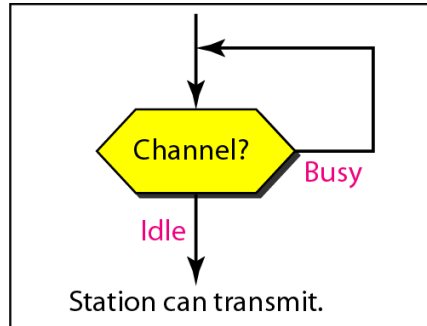
Non –persistent CSMA

- A station that has a frame to send senses the channel.
- If the channel is idle, it sends immediately.
- If the channel is busy, it waits a random amount of time and then senses the channel again.
- In non-persistent CSMA the station does not continuously sense the channel for purpose of capturing it when it detects the end of previous transmission
- Reduces the chance of collision
- Reduces the efficiency of the N/w because medium remains idle, when the data is ready to transmit

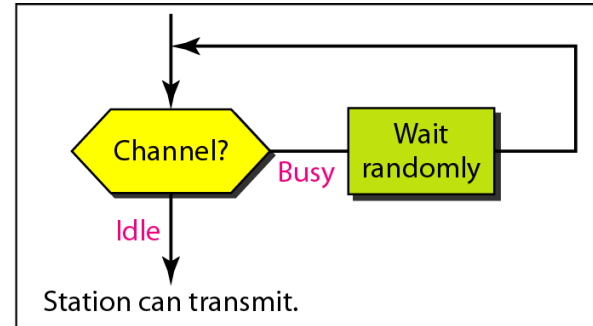
p-Persistent CSMA

- Used, if the channel has time slots with a duration equal or greater than the maximum propagation time
- Combines the advantages of other two strategies
- Reduces chance of collision and improves efficiency
 - With probability p , sends a frame
 - With prob, $q=1-p$ it checks for the line and goes to the beginning of the time slot.
 - If line is busy is assumes collision occurs and uses the back off procedure

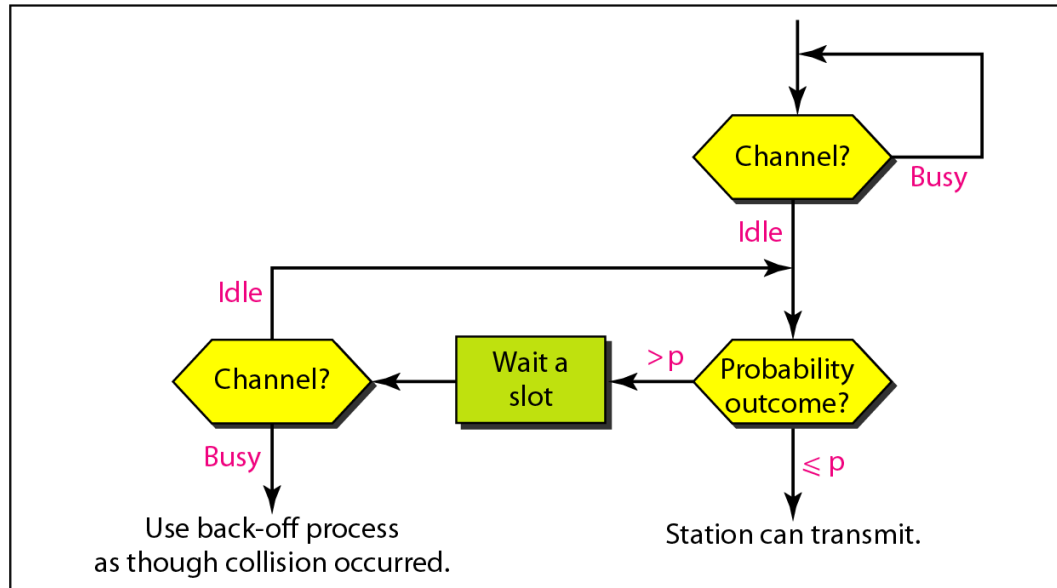
Flow diagram for three persistence methods



a. 1-persistent



b. Nonpersistent

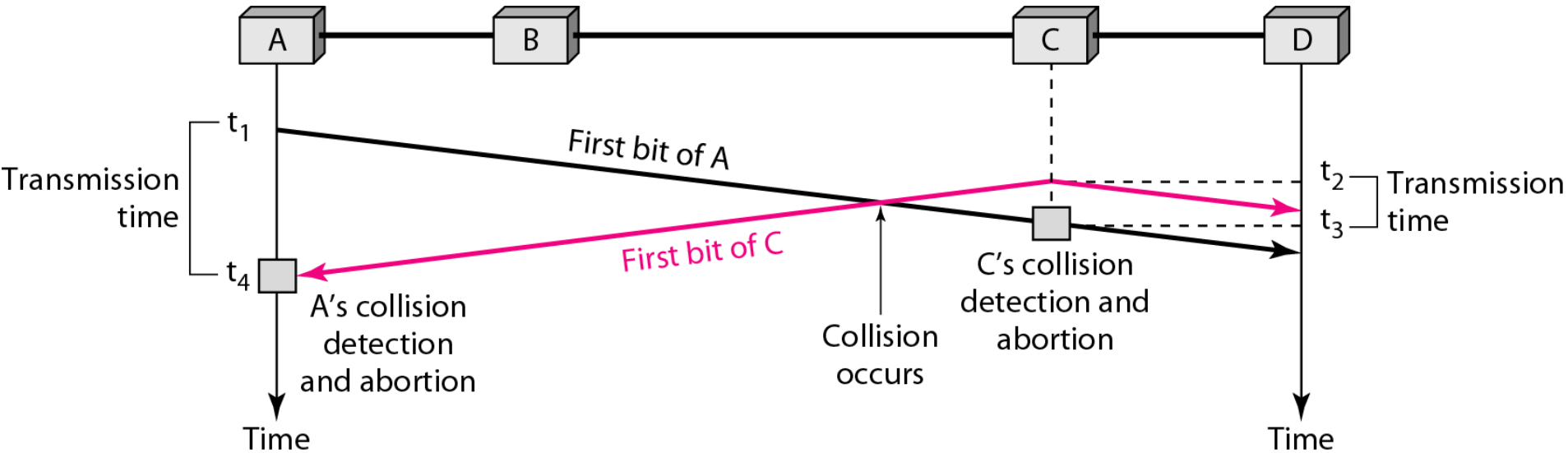


c. p-persistent

CSMA/CD

- Station monitors the medium after it transmits a frame to see if the collision occurs
 - If so, the station is finished
 - If there is a collision, frame is retransmitted
- Uses persistence method
- Transmits frame and continuously senses the medium.
- Station should stop transmission
 - If collision occurs
 - Senses collision
- Uses jamming signal to signal collision
- **Similar to ALOHA protocol** with few differences

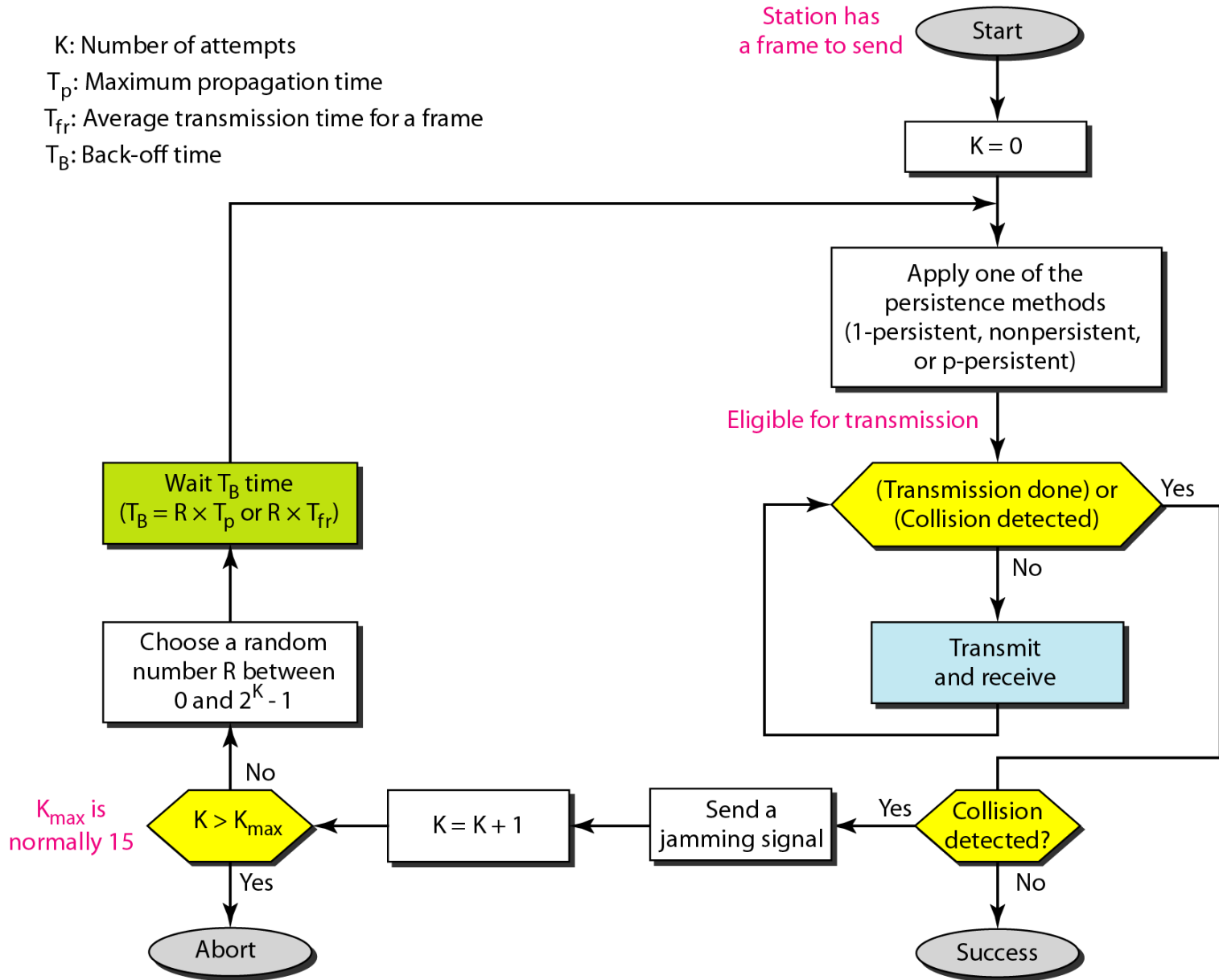
Collision of the first bit in CSMA/CD



- Having Minimum frame size or atleast equal to $2 \cdot T_p$

Flow diagram for the CSMA/CD

K: Number of attempts
 T_p : Maximum propagation time
 T_{fr} : Average transmission time for a frame
 T_B : Back-off time



Energy level during transmission, idleness, or collision

